Experiment 5 : Data Visualization using Matplotlib

Aim: To analyze Air Quality Index (AQI) trends in a city dataset and visualize the pollutant data u matplotlib plots.

In [1]:
```
# A52 Dipanshu Ambilkar
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

In [2]:
```
np.random.seed(4)
dates = pd.date_range("2026-01-01", periods=90)
df = pd.DataFrame({
    "date": dates,
    "pm25": np.random.normal(80, 25, 90).clip(10, 200),
    "pm10": np.random.normal(120, 35, 90).clip(20, 300),
    "co": np.random.normal(1.2, 0.4, 90).clip(0.2, 3),
})
df.head()
```

Out[2]:

	date	pm25	pm10	c
0	2026-01-01	81.264043	121.701105	1.288900
1	2026-01-02	92.498783	126.602617	1.621444
2	2026-01-03	55.102277	127.325972	0.946455
3	2026-01-04	97.339963	106.894428	1.051758
4	2026-01-05	69.542462	153.414451	1.014588

In [3]:
```
df["aqi"] = (df["pm25"] * 1.2 + df["pm10"] * 0.5).astype(int)
print("AQI min:", df["aqi"].min(), " AQI max:", df["aqi"].max())
```

```
AQI min: 63  AQI max: 232
```

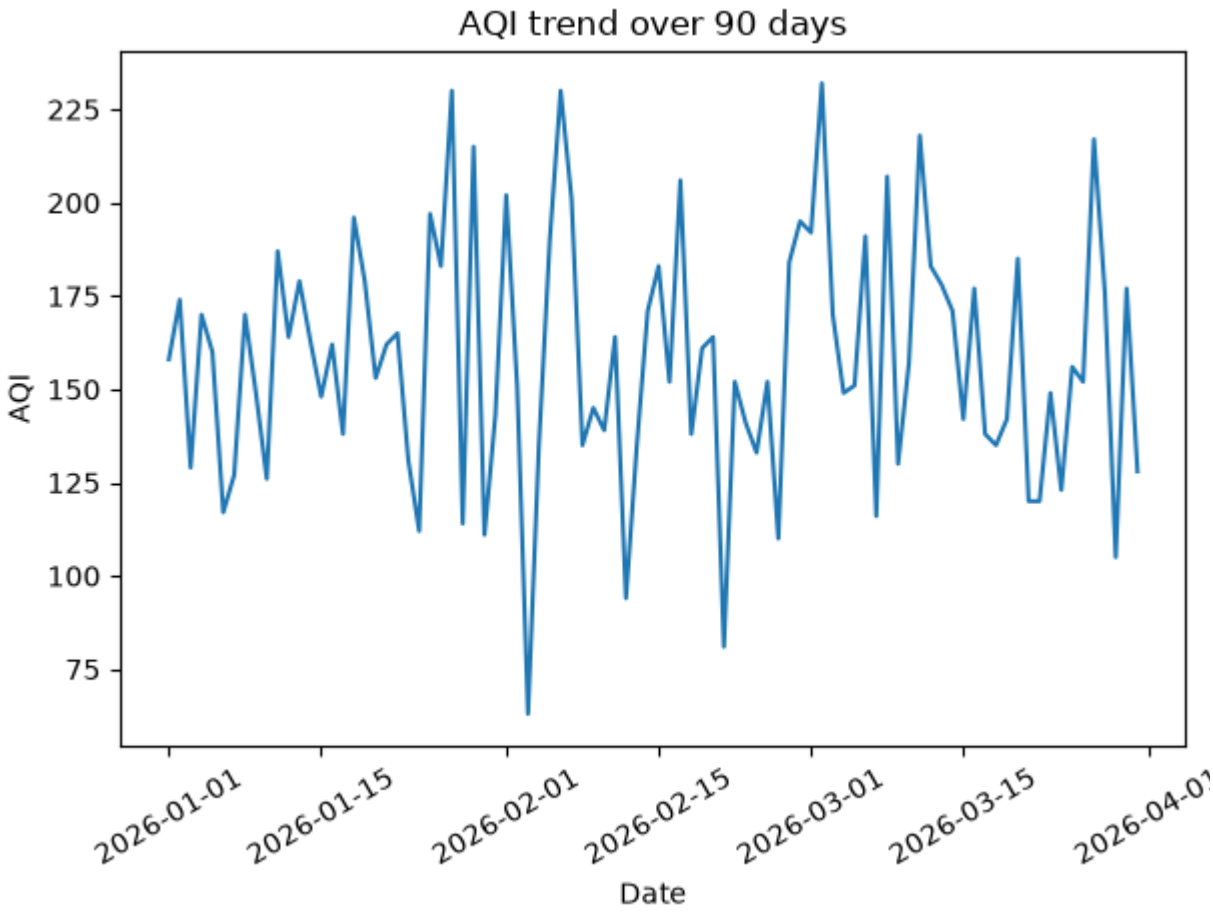
In [4]:
```
print("missing values:")
print(df.isnull().sum())
```

```
missing values:
date    0
pm25    0
pm10    0
co      0
aqi     0
dtype: int64
```

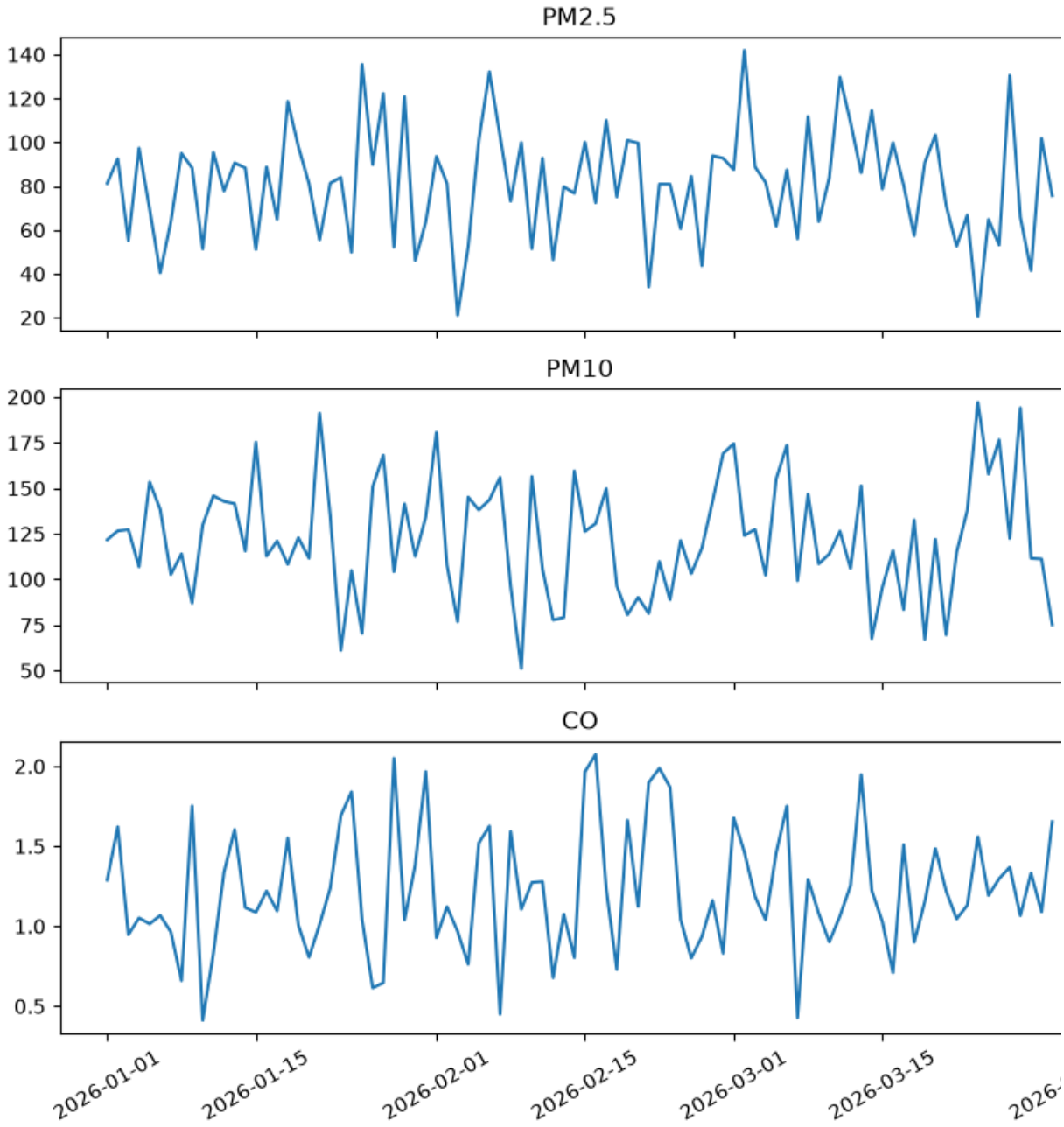
In [5]:
```
print(df.describe())
```

	date	pm25	pm10	co	aqi
count	90	90.000000	90.000000	90.000000	90.000000
mean	2026-02-14 12:00:00	80.918911	122.458523	1.221775	157.844444
min	2026-01-01 00:00:00	20.505618	51.189485	0.411299	63.000000
25%	2026-01-23 06:00:00	63.741348	103.431085	1.007870	135.000000
50%	2026-02-14 12:00:00	81.312431	121.526669	1.142707	156.500000
75%	2026-03-08 18:00:00	96.871658	143.381380	1.503508	179.000000
max	2026-03-31 00:00:00	141.914604	197.124741	2.074339	232.000000
std	NaN	25.418651	32.207356	0.388954	33.402520

In [6]:
```
plt.plot(df["date"], df["aqi"])
plt.title("AQI trend over 90 days")
plt.xlabel("Date")
plt.ylabel("AQI")
plt.xticks(rotation=30)
plt.tight_layout()
plt.show()
```



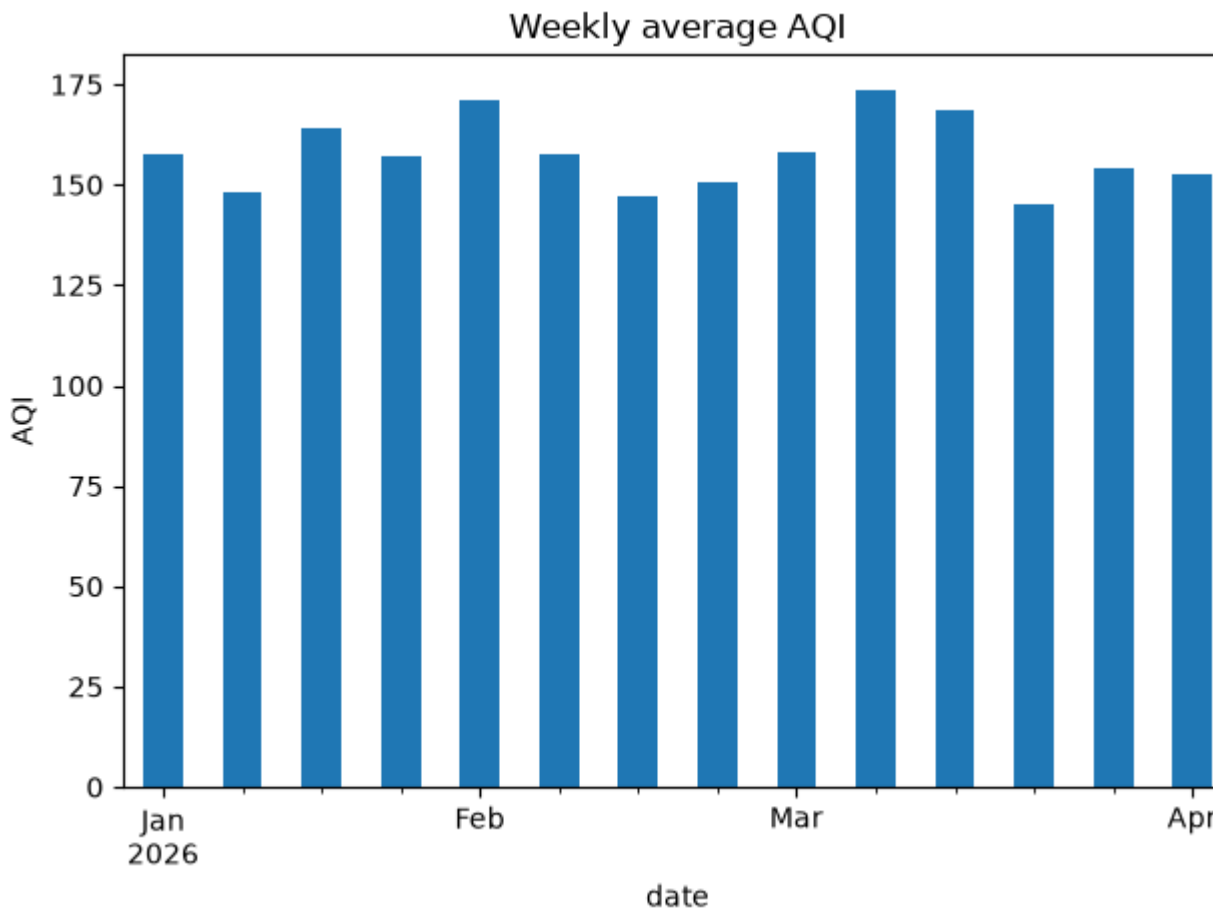
In [7]:
```
fig, ax = plt.subplots(3, 1, figsize=(8, 8), sharex=True)
ax[0].plot(df["date"], df["pm25"])
ax[0].set_title("PM2.5")
ax[1].plot(df["date"], df["pm10"])
ax[1].set_title("PM10")
ax[2].plot(df["date"], df["co"])
ax[2].set_title("CO")
plt.xticks(rotation=30)
plt.tight_layout()
plt.show()
```



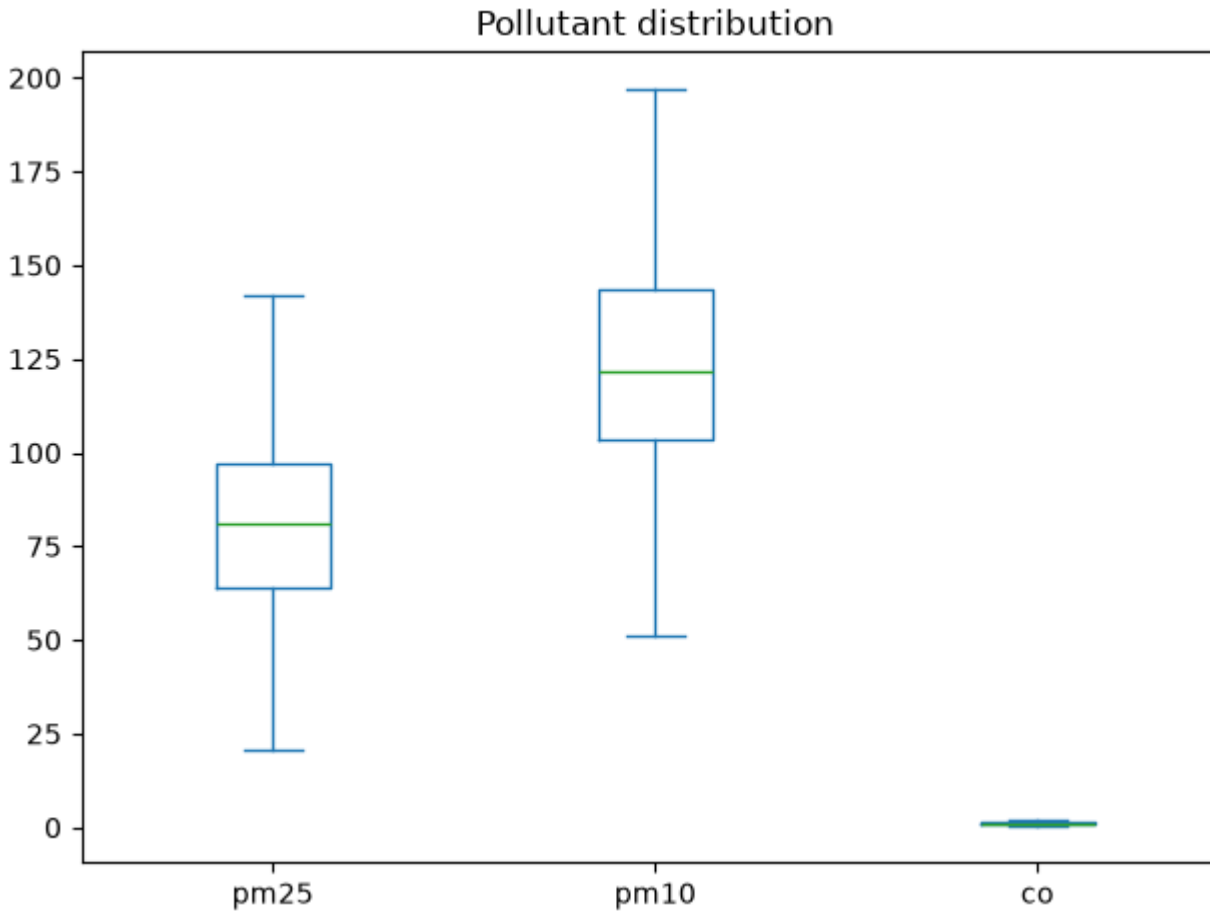
In [8]:
```
weekly = df.set_index("date")["aqi"].resample("W").mean()
print(weekly)
```

```
date
2026-01-04    157.750000
2026-01-11    148.000000
2026-01-18    164.285714
2026-01-25    157.000000
2026-02-01    171.142857
2026-02-08    157.857143
2026-02-15    147.285714
2026-02-22    150.571429
2026-03-01    158.142857
2026-03-08    173.714286
2026-03-15    168.428571
2026-03-22    145.285714
2026-03-29    154.000000
2026-04-05    152.500000
Freq: W-SUN, Name: aqi, dtype: float64
```

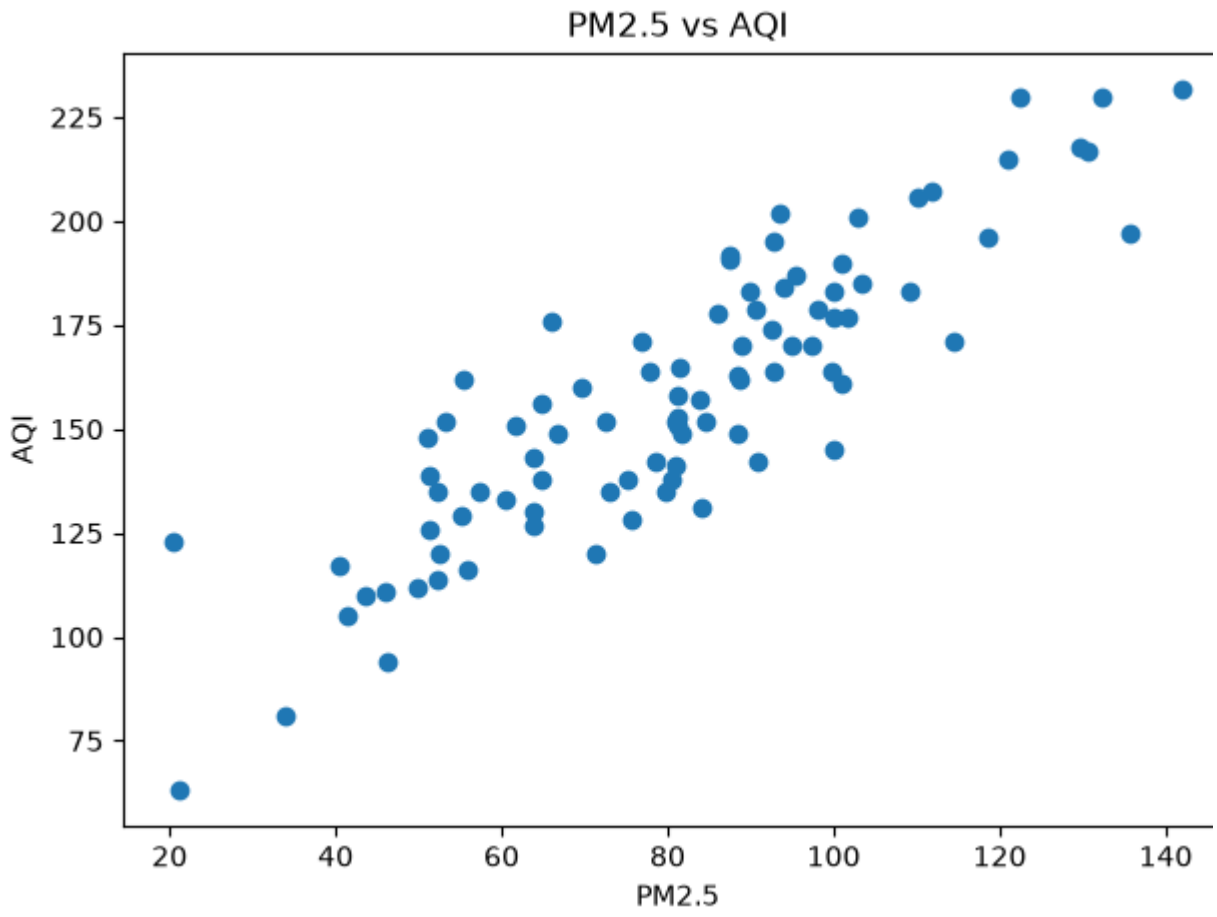
In [9]:
```
weekly.plot(kind="bar")
plt.title("Weekly average AQI")
plt.ylabel("AQI")
plt.tight_layout()
plt.show()
```



In [10]:
```
df[["pm25", "pm10", "co"]].plot(kind="box")
plt.title("Pollutant distribution")
plt.tight_layout()
plt.show()
```



In [11]:
```
plt.scatter(df["pm25"], df["aqi"])
plt.title("PM2.5 vs AQI")
plt.xlabel("PM2.5")
plt.ylabel("AQI")
plt.tight_layout()
plt.show()
```



In [12]:
```
print("done")
```

```
done
```